CS 5002 Project 1 Part 2

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[1]:
```
# ----- Import Packages -----
import numpy as np
import matplotlib.pyplot as plt
```

Section 1. Cost Function to Evaluate the Performance of a Predictive Model

1.1 Read the data given on Canvas and split into two arrays x and y

[2]:
```
def load_data(filename):
    """
    Load the data into two arrays

    Args:
        filename (str): The path to the text file containing the data (x, y)

    Return:
        array contraining the values of x
        array contraining the values of y
    """

    data = np.loadtxt(filename)
    return data[:, 0], data[:, 1]

x, y = load_data('Source/data_chol_dias_pressure.txt')
```

1.2 Write a function which computes g(a, b) for any a and b

[3]:
```
def cost_function_general (params, model_fn, x, y):
    """
    Compute the cost function for given parameters and a model function

    Args:
        params: A list or array containing the parameters such as (a, b)
        model_fn: The model function that takes parameters and x-values and␣
 ↪returns predicted y-values
        x: The x-values of the data points
        y: The y-values of the data points

    Returns:
        The cost function value
    """
    cost = 0
    for i in range(len(x)):
```

```
        cost += (model_fn(params, x[i]) - y[i]) ** 2
    return cost
```

1.3 Gradient Descent Algorithm to find optimal a and b

The optimal a and optimal b calculated from the algorithem is -4336661677168.292, -16776077493.899841 respectively.

[4]:
```
# First we can expand the previous approximate derivative functions to␣
↪vectorized form:
def approx_df(params, f, h=0.00001):
    """
    A function that implements the approximate derivative

    Args:
        params: A list or array containing the parameters such as (a, b)
        f: The cost function to compute the cost for given parameters
        h: The step size for the finite difference approximation

    Returns:
        An array containing the approximate derivatives with respect to␣
↪parmeters like (a, b)
    """
    params = np.array(params)
    n = len(params)
    df = np.zeros(n)

    for i in range(n):
        # Create a unit vector e_i with 1 at the i-th position and 0 elsewhere
        e_i = np.zeros(n)
        e_i[i] = 1
        # For the i-th parameter, compute the approximate derivative using␣
↪finite difference
        df[i] = (f(params + h * e_i) - f(params)) / h

    return df

# As we learned matrix and the way to solve using np.linalg.norm,
# we can implement the gradient descent algorithm with approximate derivatives␣
↪as follows:
def GD_approx(f, approx_df, h, params0, alpha, epsilon, iter_max=1000):
    """
    A function that implements the gradient descent algorithm with approximate␣
↪derivatives

    Args:
        f: The cost function to minimize
```

```
        approx_df: A function that computes the approximate derivative of f
        h: The step size for the finite difference approximation
        params0: Initial guess for the parameters, such as (a, b)
        alpha: Learning rate
        epsilon: Convergence threshold
        iter_max: Maximum number of iterations

    Returns:
        The optimal parameters such as (a, b) and the number of iterations␣
↪taken to converge
    """
    iter = 0
    params_current = np.array(params0)

    while (iter < iter_max):
        params_next = params_current - alpha * approx_df(params_current, f, h)

        if np.linalg.norm(params_next - params_current) < epsilon:
            return params_next, iter
        else:
            params_current = params_next
            iter += 1

    return params_next, iter
```

[5]:
```
linear_model = lambda params, x: params[0]*x + params[1]

fxy = lambda params: cost_function_general(params, linear_model, x, y)

params0 = [0.2, 40]  # Initial guess for (a, b)
param_opt, iter = GD_approx(fxy, approx_df, h=0.00001, params0=params0, alpha=0.
↪01, epsilon=1e-6)
a_opt, b_opt = param_opt
print("Optimal parameters (a*, b*):", '{:.4f}, {:.4f}'.format(a_opt, b_opt))
```

```
Optimal parameters (a*, b*): -4336661677168.2920, -16776077493.8998
```

1.4 Plot the line along with the data points of original x and y

We can see from the first plot that the plotted line is not performing as expected. **Meaning the optimal value of a and b are not ideal.**

From a rough estimation, if we assume the data points follow a **linear relation**, we should be able to fit it as $y = 0.2x + 40$. As shown in the second plot.

[6]:
```
def plot_linear_fit(x, y, a, b, title,
                    label=None,
                    xlabel='Total Cholesterol Level (mmol/L)',
```

```
                    ylabel='Diastolic Blood Pressure (mm Hg)',
                    pic_name=None):
    fig, ax = plt.subplots()
    plt.scatter(x, y)
    y_fit = a * x + b
    line_label = label if label else 'y = {:.4f}*x + {:.4f}'.format(a, b)
    plt.plot(x, y_fit, label=line_label)
    plt.legend()
    plt.xlabel(xlabel)
    plt.ylabel(ylabel)
    plt.title(title)
    fig.savefig(f'{pic_name}.png', dpi=150)
    plt.show()

plot_linear_fit(x, y, a_opt, b_opt,
                title='Fig 1: Cholesterol vs Diastolic Pressure with Optimal␣
 ↪Fit from GD',
                label='Optimal Fit, y = {:.4f}*x + {:.4f}'.format(a_opt, b_opt),
                pic_name='Linear_Fit_GD')

plot_linear_fit(x, y, 0.2, 40,
                title='Fig 2: Cholesterol vs Diastolic Pressure with Linear␣
 ↪Fit',
                label='Initial Guess, y = 0.2*x + 40',
                pic_name='Linear_Fit_Linear')
```

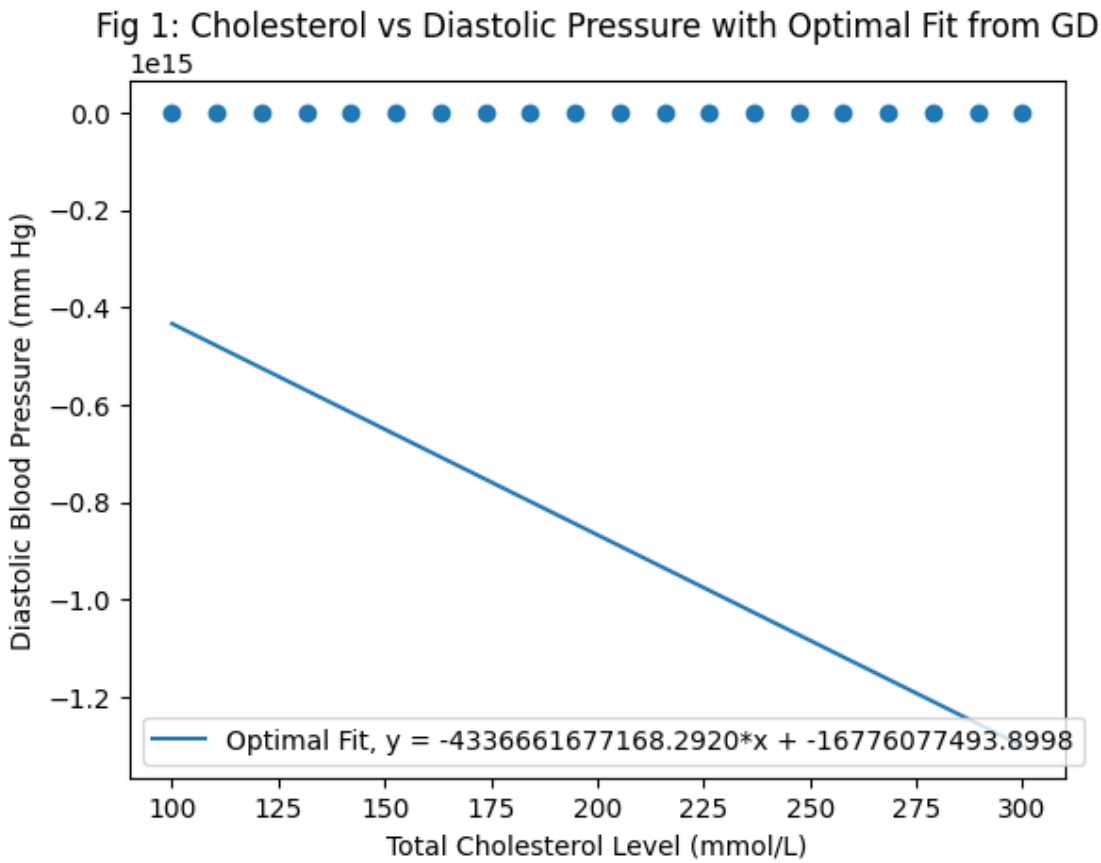
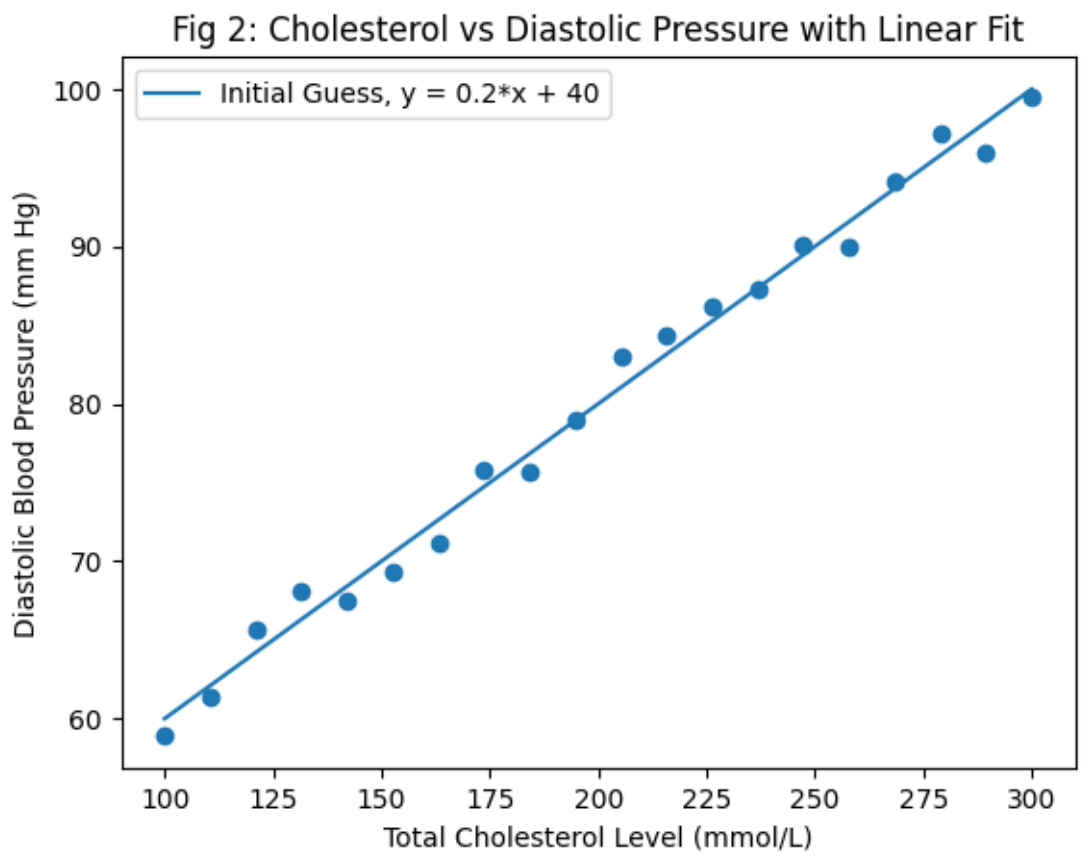


Fig 1: Cholesterol vs Diastolic Pressure with Optimal Fit from GD
1e15
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
Diastolic Blood Pressure (mm Hg)
Optimal Fit, y = -4336661677168.2920*x + -16776077493.8998
100
125
150
175
200
225
250
275
300
Total Cholesterol Level (mmol/L)



1.5 Scaling Dataset to Find the Correct Value

It was not easy to find the correct value for the intitial parameers for gradient descent to converge.

It is because before scaling, the raw x values (~200) and y values (~60-100) create **a large scale difference between parameters a and b**, causing the cost function landscape to become an elongated ellipse rather than a symmetric bowl. This makes gradient descent unstable — any single learning rate is either too large (overshooting in the steep direction) or too small (crawling in the flat direction). Standardizing the data brings all values to a similar scale (~0 to ±2), reshaping the landscape into a well-conditioned bowl that gradient descent can navigate efficiently regardless of the model used.

After scaling, we are able to find the correct optimal value for a and b as

$$a^* = 0.1998 \quad \text{and} \quad b^* = 40.0521$$

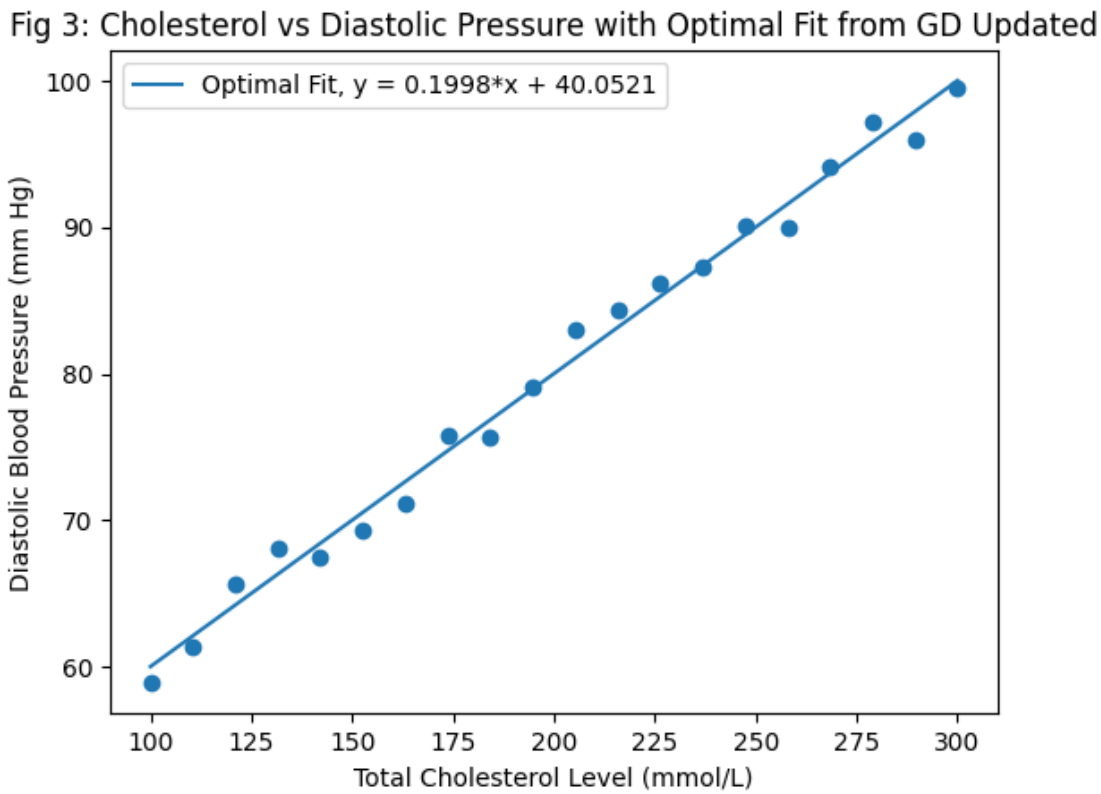
And the fitting line looks good now with the updated values as in plot 3.

[7]:
```
# First we can scale the data to have zero mean and unit variance,
# which can help improve the convergence of gradient descent:
def scale_data(x, y):
```

```python
    """
    Scale the data to have zero mean and unit variance

    Args:
        x: The x-values of the data points
        y: The y-values of the data points

    Returns:
        Scaled x and y values
    """

    x_scaled = (x - np.mean(x)) / np.std(x)
    y_scaled = (y - np.mean(y)) / np.std(y)
    return x_scaled, y_scaled

# After we get the optimal parameters in the scaled space,
# we can transform them back to the original space using the following function:
def transform_params(params, x, y):
    """
    Transform the parameters from the scaled space back to the original space

    Args:
        params: A list or array containing the parameters such as (a, b) in the␣
 ↪scaled space
        x: The original x-values of the data points
        y: The original y-values of the data points

    Returns:
        Transformed parameters in the original space
    """
    a_scaled, b_scaled = params
    a_original = a_scaled * (np.std(y) / np.std(x))
    b_original = np.mean(y) - a_original * np.mean(x)
    return a_original, b_original


x_scaled, y_scaled = scale_data(x, y)
linear_model = lambda params, x: params[0]*x + params[1]
fxy = lambda params: cost_function_general(params, linear_model, x_scaled,␣
 ↪y_scaled)

params0 = [0.2, 40]  # Initial guess for (a, b)
param_opt, iter = GD_approx(fxy, approx_df, h=0.00001, params0=params0, alpha=0.
 ↪01, epsilon=1e-6)
a_opt, b_opt = transform_params(param_opt, x, y)
print("Optimal parameters (a*, b*) after transformation:", '{:.4f}, {:.4f}'.
 ↪format(a_opt, b_opt))
```

```
plot_linear_fit(x, y, a_opt, b_opt,
                title='Fig 3: Cholesterol vs Diastolic Pressure with Optimal␣
 ↪Fit from GD Updated',
                label='Optimal Fit, y = {:.4f}*x + {:.4f}'.format(a_opt, b_opt),
                pic_name='Linear_Fit_GD_Updated')
```

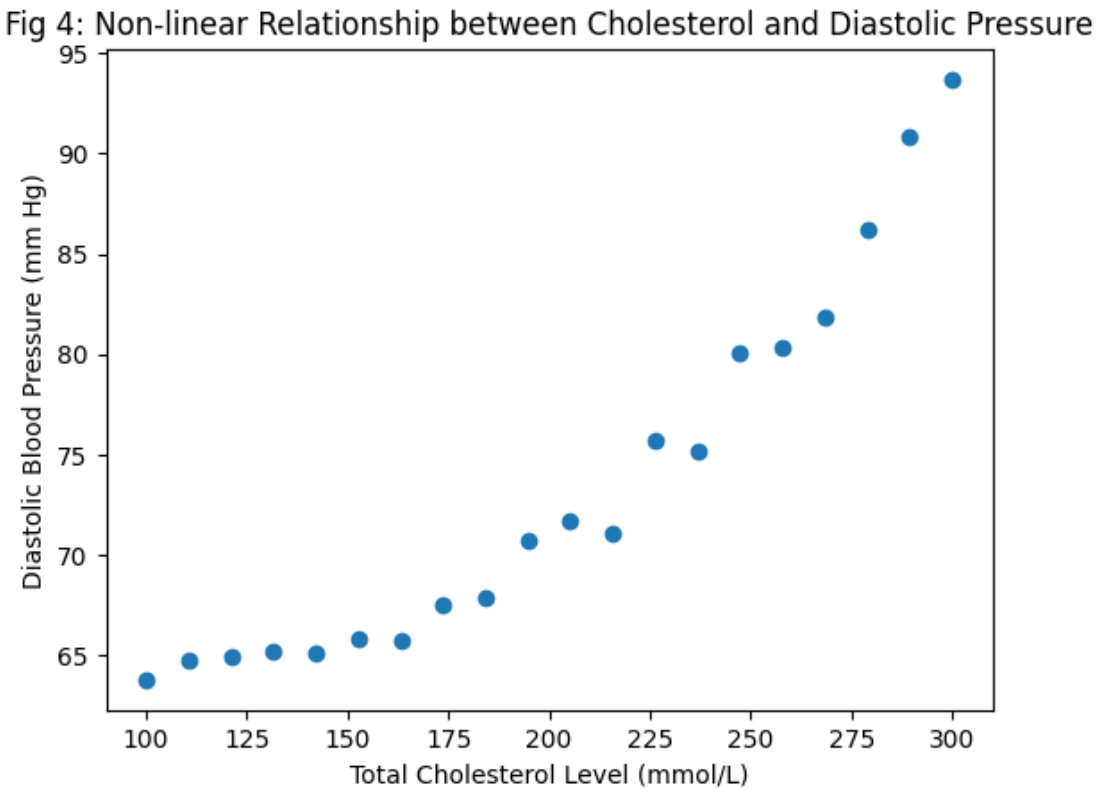
```
Optimal parameters (a*, b*) after transformation: 0.1998, 40.0521
```



Section 2. Non-linear Data

2.1 Read the data given on Canvas and split into two arrays x and y

[8]:
```
x_non_lin, y_non_lin = load_data('Source/data_chol_dias_pressure_non_lin.txt')
plt.scatter(x_non_lin, y_non_lin)
plt.xlabel('Total Cholesterol Level (mmol/L)')
plt.ylabel('Diastolic Blood Pressure (mm Hg)')
plt.title('Fig 4: Non-linear Relationship between Cholesterol and Diastolic␣
 ↪Pressure')
plt.savefig('Non_Linear_Relationship.png', dpi=150)
plt.show()
```



2.2 Apply the Gradient Descent algorithm and get the optimal a and b

By applying the GD algorithm, we find the optimal value for a and b as

$$a^* = 0.1400 \quad \text{and} \quad b^* = 45.3867$$

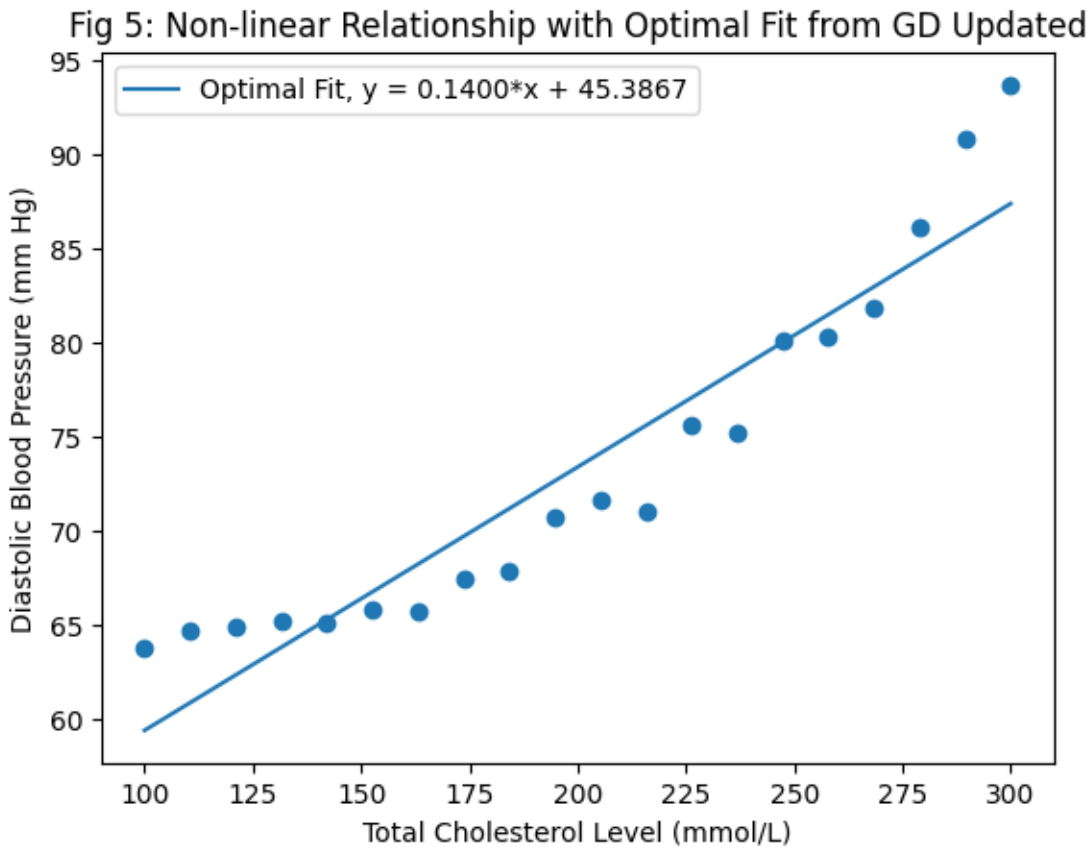
[9]:
```
x_scaled, y_scaled = scale_data(x_non_lin, y_non_lin)
linear_model = lambda params, x: params[0]*x + params[1]
fxy = lambda params: cost_function_general(params, linear_model, x_scaled,␣
↪y_scaled)

params0 = [0.2, 40]  # Initial guess for (a, b)
param_opt, iter = GD_approx(fxy, approx_df, h=0.00001, params0=params0, alpha=0.
↪01, epsilon=1e-6)
a_opt, b_opt = transform_params(param_opt, x_non_lin, y_non_lin)
print("Optimal parameters (a*, b*) after transformation:", '{:.4f}, {:.4f}'.
↪format(a_opt, b_opt))
```

```
Optimal parameters (a*, b*) after transformation: 0.1400, 45.3867
```

2.3 Plot the line and see how it perform

The plotted line perform not very well. This is because we can see from the chart that the data is not linear, so a linear line $y = a^*x + b^*$ will not show a good fitting.

[10]:
```
plot_linear_fit(x_non_lin, y_non_lin, a_opt, b_opt,
                title='Fig 5: Non-linear Relationship with Optimal Fit from GD␣
 ↪Updated',
                label='Optimal Fit, y = {:.4f}*x + {:.4f}'.format(a_opt, b_opt),
                pic_name='Non_Linear_Fit_GD_Updated')
```



2.4 Possible modifications to improve its performance - Consider non-linear model

By looking at the scatter plot of the non-linear data, we can see that the data points do not follow a linear pattern.

We can consider non-linear models. Since the scatter plot seems to follow a quadratic model, try:

$$f(x) = ax^2 + bx + c$$

As we can see from Fig 6, quadratic model fits well for this non-linear data set with a function as:

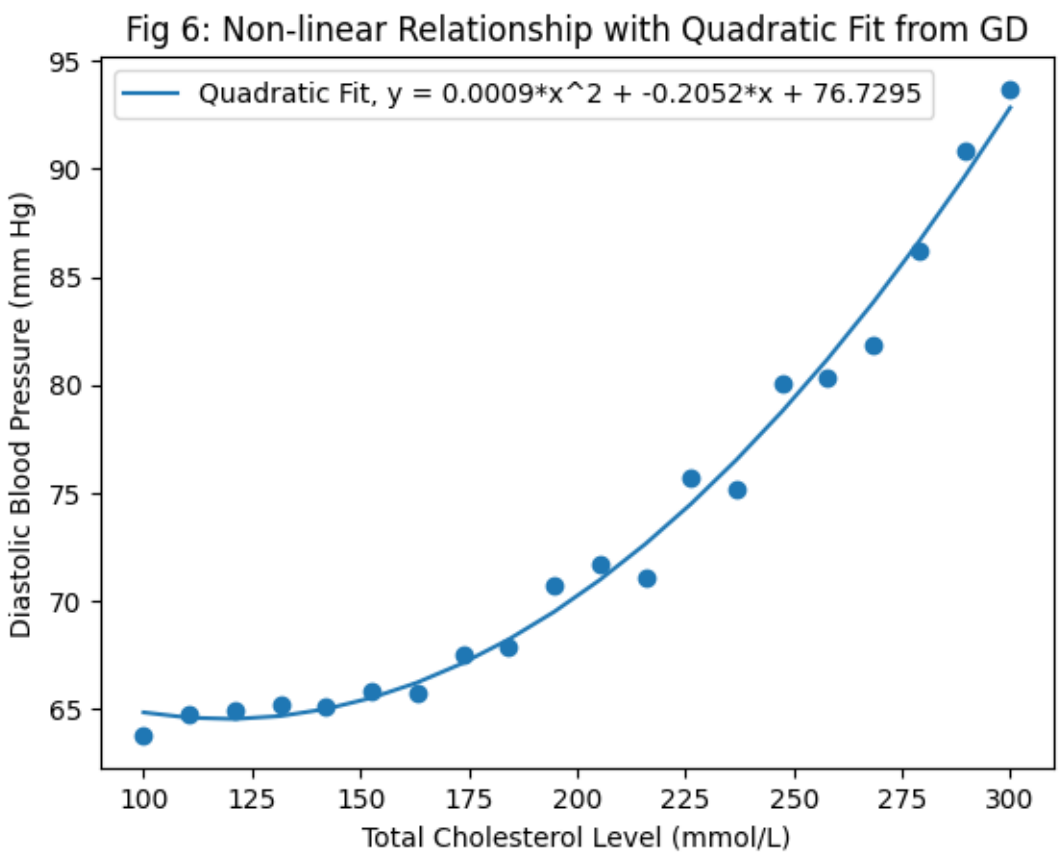
$$y = 0.0009x^2 - 0.2052x + 76.7295$$

[11]:
```python
# Some helper functions for the non-linear model fitting:

# Transform the parameters back to the original space
def transform_quad_params(params, x, y):
    """
    Transform quadratic model parameters from scaled (x_s, y_s) space back to
    original (x, y) space, where x_s = (x -  _x)/_x and y_s = (y -  _y)/_y.

    Scaled model:    y_s = a_s * x_s^2 + b_s * x_s + c_s
    Original model:  y   = A * x^2 + B * x + C
    """
    a_s, b_s, c_s = params
    mu_x, sigma_x = np.mean(x), np.std(x)
    mu_y, sigma_y = np.mean(y), np.std(y)

    A = a_s * sigma_y / (sigma_x ** 2)
    B = (b_s * sigma_y / sigma_x) - (2 * a_s * sigma_y * mu_x / (sigma_x ** 2))
    C = (a_s * sigma_y * (mu_x ** 2) / (sigma_x ** 2)
         - b_s * sigma_y * mu_x / sigma_x
         + c_s * sigma_y
         + mu_y)
    return A, B, C

# A function to plot the quadratic fit
def plot_quadratic_fit(x, y, a, b, c, title,
                       label=None,
                       xlabel='Total Cholesterol Level (mmol/L)',
                       ylabel='Diastolic Blood Pressure (mm Hg)',
                       pic_name=None):
    plt.scatter(x, y)
    y_fit = a * x**2 + b * x + c
    line_label = label if label else 'y = {:.4f}*x^2 + {:.4f}*x + {:.4f}'.
 ↪format(a, b, c)
    plt.plot(x, y_fit, label=line_label)
    plt.legend()
    plt.xlabel(xlabel)
    plt.ylabel(ylabel)
    plt.title(title)
    plt.savefig(pic_name, dpi=150)
    plt.show()
```

[12]:
```python
quadratic_model = lambda params, x: params[0]*x**2 + params[1]*x + params[2]
fxy_quad = lambda params: cost_function_general(params, quadratic_model,␣
 ↪x_scaled, y_scaled)

params0_quad = [0.0, 0.0, 0.0]  # Initial guess for (a, b, c)
```

```
param_opt_quad, iter_quad = GD_approx(fxy_quad, approx_df,
                                      h=0.00001,
                                      params0=params0_quad,
                                      alpha=0.01,
                                      epsilon=1e-6,
                                      iter_max=1000)

a_opt_quad, b_opt_quad, c_opt_quad = transform_quad_params(param_opt_quad,␣
 ↪x_non_lin, y_non_lin)
print("Optimal parameters (a*, b*, c*) for quadratic model after transformation:
 ↪",
      '{:.4f}, {:.4f}, {:.4f}'.format(a_opt_quad, b_opt_quad, c_opt_quad))

plot_quadratic_fit(x_non_lin, y_non_lin, a_opt_quad, b_opt_quad, c_opt_quad,
                   title='Fig 6: Non-linear Relationship with Quadratic Fit␣
 ↪from GD',
                   label='Quadratic Fit, y = {:.4f}*x^2 + {:.4f}*x + {:.4f}'
                   .format(a_opt_quad, b_opt_quad, c_opt_quad),
                   pic_name='Non_Linear_Fit_Quadratic_GD')
```

```
Optimal parameters (a*, b*, c*) for quadratic model after transformation:
0.0009, -0.2052, 76.7295
```



Section 3. Report

1. Designing Gradient Descent for Functions of Several Variables

To minimize the cost function `g(a, b)` (and later `g(a, b, c)` for the quadratic model), I generalized the single-variable gradient descent algorithm to operate on a parameter *vector*. Three design choices made this generalization clean and reusable:

1.1 Vectorized Approximate Derivative

The function `approx_df` computes the partial derivative of the cost with respect to each parameter using a finite-difference approximation along the unit vector `e_i`:

$$\frac{\partial g}{\partial \theta_i} \approx \frac{g(\theta + h \cdot e_i) - g(\theta)}{h}$$

Looping over each parameter and assembling the results into an array produces the full gradient vector. This works for any number of parameters, make it able to use *same* `approx_df` for both the 2-parameter linear model and the 3-parameter quadratic model without modification.

1.2 Generic Cost Function

Cost function `cost_function_general(params, model_fn, x, y)` is updated to accept any model function as an argument. This made it trivial to swap models — switching from `linear_model` to `quadratic_model` required only changing one argument, while the GD machinery stayed identical.

1.3 Vector Convergence Check

The 1D convergence criterion `|x_next - x_current| <` generalizes naturally to vectors:

$$\|\theta_{\text{next}} - \theta_{\text{current}}\| < \varepsilon$$

From class we learned the package np.linalg, and `np.linalg.norm` is useful to measures the overall magnitude of the parameter update across all dimensions.

2. Challenges Encountered and Solutions

2.1 Sensitivity to Initial Parameters and Learning Rate

When applying GD directly to the raw data, convergence was extremely sensitive to the choice of `params0` and `alpha`, and wasn't able to find the correct optimal easily.

Why this happened: With cholesterol values in the range 100–300 mmol/L, the cost surface is severely ill-conditioned. The gradient along the `a` direction (involving `x²` 90,000) is many orders of magnitude larger than the gradient along the intercept direction.

2.2 Solution: Data Scaling

Following the hint in Question 1.5, scaling both `x` and `y` to have mean 0 and standard deviation 1.

After scaling, all parameters operate on similar magnitudes, the cost surface becomes well-conditioned.

2.3 Transforming Parameters Back to Original Units

A subtle challenge was converting scaled-space parameters back to original-space coefficients for reporting and plotting. For the quadratic model:

$$y_s = a_s x_s^2 + b_s x_s + c_s$$

substituting $x_s = (x - \mu_x)/\sigma_x$ and $y_s = (y - \mu_y)/\sigma_y$ and expanding the squared term:

$$\left(\frac{x - \mu_x}{\sigma_x}\right)^2 = \frac{x^2 - 2\mu_x x + \mu_x^2}{\sigma_x^2}$$

With Claude's hint, I was able to get the transform correctly, but it took me a long time to get the expressions correctly.

3. Effect of Model Choice on Performance

The two datasets demonstrated clearly that **gradient descent is only as good as the model it optimizes**.

3.1 Dataset 1 — Linear Relationship

The first dataset showed a clean linear trend. The model `f(x) = ax + b` fit the data well after scaling, with the optimal line passing tightly through the data cloud. Both the model form and the optimization procedure were appropriate.

3.2 Dataset 2 — Non-Linear Relationship

The new dataset exhibited upward curve: blood pressure was nearly flat at low cholesterol levels but accelerated rapidly at higher levels. Applying the **same linear model** to this data produced a noticeably worse fit — the line systematically mispredicted both the low and high ends of the cholesterol range.

3.3 Switching to a Quadratic Model

Replacing `f(x) = ax + b` with `f(x) = ax² + bx + c`, after applying the scaling and transformation procedure described in Section 2, the fitting quadratic model is:

$$y = 0.0009\,x^2 - 0.2052\,x + 76.7295$$

The updated fitting line works much better as seen in Figure 6.

3.4 Key Insight

Optimization and modeling are **separate concerns**, and both must be addressed for a useful predictive model:

- **Modeling** chooses the *space of possible relationships* the model can represent (linear, quadratic, power law, etc.). This decision must come *before* optimization.
- **Optimization** finds the best parameters *within that space*. GD cannot rescue a model whose functional form is too restrictive for the data.

A more expressive model (more parameters) can capture more patterns but also makes the cost landscape harder to navigate, increasing the importance of scaling and good initialization. There is a trade-off between expressiveness and trainability that good machine learning practice must balance.

4. Lessons Learned

1. **Generalizing GD to multiple variables is a clean abstraction.** Once the gradient is treated as a vector and convergence is measured by norm, the algorithm extends to any number of parameters with no code changes.
2. **Data scaling is essential.** Without scaling, the cost surface for raw cholesterol values is too ill-conditioned for vanilla GD to navigate efficiently.
3. **Parameter transformations require care.** Substituting scaled variables back into the model introduces algebraic cross-terms that are easy to miss, and getting them wrong can mimic the appearance of optimization failure.
4. **Debugging GD requires checking multiple hypotheses.** When a fit looks bad, the cause may be: insufficient iterations, a poor learning rate, missing data scaling, an incorrect parameter transformation, or even an inadequate model. I encountered each of these at different points in this project.
5. **The choice of `f(x)` defines the ceiling of model performance.** No optimizer can exceed what the model form allows. Selecting an appropriate functional form for the data is the first and most consequential modeling decision.